

RESEARCH ARTICLE

Predictive Modeling of Solar Energy Output in The Gambia Using Machine Learning: A Comparative Study of ML Algorithms Under Tropical West African Climate Conditions

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Abstract

Accurate prediction of solar energy output is essential for the reliable integration of photovoltaic (PV) systems into national electricity grids, particularly in sub-Saharan Africa where challenges of energy access persist despite exceptional solar resources. This study presents the first comprehensive machine learning (ML) - based framework for predicting solar irradiance and photovoltaic energy output specifically tailored to The Gambia. Note that The Gambia is a tropical West African nation with mean annual global horizontal irradiance (GHI) of approximately 5.8 kWh/m²/day. Five Machine Learning(ML) algorithms were developed and compared: Multiple Linear Regression (MLR), Support Vector Regression (SVR), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Long Short-Term Memory (LSTM) neural networks. The models were trained on twenty-two years (2000–2022) of hourly meteorological data sourced from NASA POWER and ERA5 reanalysis datasets, incorporating ten input features including GHI, clearness index, air temperature, relative humidity, aerosol optical depth, wind speed, and temporal variables. Model performance was evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2). The LSTM model achieved the best overall performance with RMSE = 48.2 W/m², MAE = 33.1 W/m², MAPE = 7.6%, and R^2 = 0.963. Seasonal disaggregation revealed that the harmattan dust period (December–March) introduced the highest prediction uncertainty, with RMSE increasing to 68.3 W/m², attributed to aerosol optical depth variability. SHAP (SHapley Additive exPlanations) analysis identified GHI, clearness index, and air temperature as the three most influential features. The findings provide actionable insights for national utility operators, off-grid system designers, and energy policymakers in The Gambia and comparable West African nations.

1. Introduction

1.1 Background and Motivation

The global transition toward renewable energy has accelerated sharply over the past decade, with solar photovoltaic (PV) capacity exceeding 1.2 terawatts worldwide by 2023 (IRENA 2023). Sub-Saharan Africa possesses some of the world's highest

solar irradiance levels, with daily Global Horizontal Irradiance(GHI) values consistently ranging from 4.5 to 7.0 kWh/m² across the region, yet it accounts for less than 2% of global installed solar capacity (Africa Energy Outlook 2022). This paradox of abundant solar resource coupled with low utilization, is driven by multiple barriers including limited technical expertise, financing constraints, weak grid infrastructure, and crucially, the absence of reliable localized solar energy prediction systems.

The Gambia, the smallest country on the African mainland, exemplifies this challenge. Situated between 13.0°N and 13.8°N latitude along the Atlantic coast of West Africa, the country receives mean annual GHI of approximately 5.5–6.0 kWh/m²/day, placing it among the most solar-rich nations globally (Stackhouse, P.W, et al, 2018). The national grid, operated by the National Water and Electricity Company (NAWEC), suffers from chronic power deficits, with the national electrification rate standing at approximately 63% as of 2022 (World Bank 2023). Frequent load-shedding, unreliable supply, and the high cost of diesel-generated electricity create a compelling case for accelerated solar deployment.

Despite these favorable conditions, solar energy contributes only a minor fraction of The Gambia's electricity mix. A critical technical barrier to increasing solar penetration in the national grid is the variability of solar output caused by cloud cover, humidity, and seasonal dust from the harmattan .During this period, a dry, dusty trade wind that blows from the Sahara across West Africa between November and March, deposite aerosols that significantly attenuate solar radiation. Accurate prediction of solar output under these conditions is essential for dispatch planning, battery storage sizing, and economic assessment of solar investments [Diagne,M., David, M., Lauret, P., Boland, J., and Schmutz 2013).

Machine learning has emerged as a powerful tool for solar energy forecasting because it can capture complex nonlinear relationships between meteorological variables and solar output without requiring explicit physical modeling of atmospheric processes (Voyant, Notton et al 2017). However, almost all existing ML-based solar forecasting studies have focused on Europe, North America, and Asia, with very limited coverage of West Africa and virtually no studies targeting The Gambia at the national level.

1.2 Problem Statement

The core problem addressed in this study is the absence of accurate, locally calibrated solar energy prediction models for The Gambia. Generic global models or continental-scale models trained predominantly on temperate-zone data perform poorly when applied to The Gambia. This is because they fail to capture the specific influence of harmattan aerosols, the sharp dry/wet season transition, and the relatively high humidity of the coastal belt. Furthermore, the sparse network of ground-based meteorological stations in The Gambia means that any operational forecasting system must effectively utilize satellite-derived and reanalysis data products. Without reliable solar output forecasts, grid operators cannot efficiently dispatch generation, investors cannot accurately project revenue, and policymakers lack the technical evidence base needed to set feed-in tariffs and incentive structures. The absence of prediction capability is therefore not merely an academic gap but a practical barrier to solar energy deployment.

1.3 Research Objectives

This study pursues four specific objectives:

- To develop and rigorously compare five Machine Learning algorithms — MLR, SVR, Random Forest, XGBoost, and LSTM — for predicting solar irradiance and PV output in The Gambia using 22 years of hourly reanalysis data.
- To conduct seasonal disaggregation of model performance, distinguishing between the dry season (November–April), the wet season (May–October), and the harmattan dust period (December–March) as a sub-season of particular predictive difficulty.
- To apply SHAP explainability analysis to identify the most influential meteorological and derived features driving prediction accuracy.
- To translate modeling findings into actionable policy and engineering recommendations relevant to NAWEC, off-grid system designers, and national energy planners in The Gambia and comparable West African nations.

1.4 Contributions and Paper Structure

The principal contributions of this work are: (i) the first ML-based solar energy prediction study explicitly focused on The Gambia, filling a clear gap in the African solar forecasting literature; (ii) a systematic multi-model comparison under tropical West African climate conditions, providing reproducible benchmarks for future researchers; (iii) a harmattan-specific seasonal analysis quantifying the impact of dust aerosols on prediction accuracy; and (iv) an explainability analysis that reveals

which physical variables are most important for localized forecasting, with practical implications for sensor investment priorities. The remainder of this paper is organized as follows: Section 2 reviews the relevant literature on ML-based solar forecasting, with emphasis on African and tropical applications. Section 3 describes the study area and data sources. Section 4 presents the methodology, including feature engineering, model architectures, and evaluation metrics. Section 5 reports results and discussion. Section 6 outlines practical implications for energy planning in The Gambia. Section 7 concludes the paper with limitations and directions for future research.

2. Literature Review

2.1 Machine Learning Approaches in Solar Energy Forecasting

The application of machine learning to solar energy forecasting has grown substantially over the past decade. Early studies relied primarily on Artificial Neural Networks (ANNs) and Support Vector Regression (SVR) to capture the nonlinear relationship between atmospheric variables and solar irradiance. Yadav and Chandel in 2014 conducted a comprehensive review of 162 ANN-based solar radiation models, finding R^2 values consistently above 0.90 for well-trained networks, though noting that performance degraded significantly when models were applied to geographic contexts different from their training data (Yadav, Chandel 2014).

Ensemble methods, particularly Random Forest and Gradient Boosting variants, have more recently demonstrated competitive or superior performance to neural networks in many forecasting contexts. Abdel-Nasser and Mahmoud compared Random Forest, Extreme Learning Machine, and SVR for daily solar irradiance forecasting and found that ensemble tree methods achieved lower RMSE in most test scenarios, with the additional advantage of built-in feature importance estimation (Abdel-Nasser, Mahmoud 2014). In addition to that, XGBoost, introduced by Chen and Guestrin, has proven especially effective for tabular meteorological data due to its regularization framework and efficient handling of feature interactions, achieving state-of-the-art performance in several benchmark studies (Chen, Guestrin, et al 2016). Deep learning approaches, particularly Long Short-Term Memory (LSTM) networks, have gained prominence for time-series solar forecasting because they can explicitly model temporal dependencies across multiple time steps, capturing

diurnal patterns and multi-day weather sequences that tree-based models treat as independent observations (Agga, Abbou, et al 2021). Abdeljaber et al in 2017 demonstrated that LSTM outperformed both SVR and Random Forest for hourly GHI forecasting across multiple Middle Eastern sites, with improvements in RMSE of 15–23% compared to the best non-sequential baselines. Hybrid architectures combining convolutional layers with LSTM have also shown promise, though their greater computational demands and training complexity make them less practical for resource-constrained deployments.

2.2 Solar Forecasting in Africa and Tropical Climates

Despite the urgency of solar energy expansion across Africa, the continent remains underrepresented in the solar forecasting literature. Voyant et al in 2017 noted in their research on “Solar Resource Assessment In West Africa Using Satellite Derived Data Renewable Energy Focus”, that fewer than 3% of solar forecasting studies in a comprehensive 2017 review covered African sites. More recent work has begun to address this gap, but the coverage remains skewed toward North Africa (particularly Egypt, Morocco, and Algeria) and South Africa, with West and Central Africa receiving very limited attention (Ouedraogo, Coulibaly, Tigabu, et al).

Studies in tropical and Sahelian climates highlight several forecasting challenges that differ from temperate-zone contexts. Diagne et al in 2014 found that cloud variability in humid tropical zones introduced higher forecast errors compared to Mediterranean sites, with MAPE values exceeding 15% during the rainy season even with well-calibrated models. Salima and Chavula investigated solar radiation estimation in Malawi using various empirical and regression models and found that models calibrated for East African conditions could not be directly transferred to Central African sites without significant recalibration (Salima and Chavula 2012). This spatial heterogeneity of model performance underscores the need for locally calibrated models rather than the application of generic global models.

Aerosol optical depth (AOD) which is the primary optical measure of atmospheric dust loading, has been identified as a critical and often neglected input variable for West African solar forecasting. Ndoeye et al in 2019 demonstrated that harmattan dust events in Senegal could reduce daily GHI by 30–45% compared to clear-sky values, with significant day-to-day variability that makes persistence forecasting

unreliable during this period. Including AOD as a model input substantially improved prediction accuracy, reducing RMSE by 18–25% during dust-affected days in their study.

2.3 Key Input Features and Data Sources

The selection of input features is a critical determinant of solar forecasting model performance. Lorenz et al in 2009, found that models incorporating at least GHI, temperature, relative humidity, and cloud cover metrics consistently outperformed models using only one or two variables. The clearness index (K_t), defined as the ratio of measured GHI to extraterrestrial irradiance on a horizontal surface, has emerged as one of the most widely used derived features because it normalizes for the solar geometry component and isolates the atmospheric transmittance signal (Duffie, Beckman and Blair 2009).

For regions with sparse ground station networks (which characterizes virtually all of sub-Saharan Africa), satellite-derived and reanalysis data products are the primary data sources. The NASA Prediction of Worldwide Energy Resources (POWER) database provides 40+ years of daily and hourly surface meteorological and solar energy data at $0.5^\circ \times 0.5^\circ$ spatial resolution, derived from the MERRA-2 reanalysis, and has been validated for African applications by Stackhouse et al. The ERA5 reanalysis from the European Centre for Medium-Range Weather Forecasts (ECMWF) provides higher spatial and temporal resolution (31 km, hourly) and includes aerosol optical depth variables, making it particularly valuable for dust-affected forecasting contexts.

2.4 Research Gaps

The review above reveals three clear research gaps that this study addresses. First, no published study has developed or validated ML-based solar energy prediction models specifically for The Gambia. The most geographically proximate published work covers Senegal (approximately 400 km north) and Mali (inland, with distinctly different climate), leaving a gap that is both geographically and climatologically meaningful. Second, existing African solar forecasting studies rarely conduct rigorous seasonal disaggregation, and none to the authors' knowledge have specifically isolated and analyzed harmattan period performance as a distinct predictive challenge requiring specialized treatment. Third, explainability analysis

using SHAP or comparable methods has been applied to solar forecasting in Asian and European contexts but not in West African case studies, limiting the transferability of future importance findings to this regional context.

3. Study Area and Data

3.1 Study Area: The Gambia

The Gambia is a narrow, elongated country in West Africa, straddling the Gambia River from the Atlantic coast inland for approximately 480 km. It covers an area of approximately 11,300 km² and is entirely surrounded by Senegal except for its Atlantic coastline. The country lies between latitudes 13.0°N–13.8°N and longitudes 13.8°W–16.9°W. Its small geographic extent and relative topographic uniformity (maximum elevation below 60 m), mean that a single site model can be reasonably representative of national level conditions, though coastal-inland gradients in humidity and cloud frequency are noted.

The climate is tropical with a distinct seasonality governed primarily by the migration of the Inter-Tropical Convergence Zone (ITCZ). The dry season extends from approximately November to April, characterized by low humidity, clear skies, and the prevalence of harmattan winds carrying mineral dust from the Sahara. Mean daily GHI during the dry season reaches 6.2–6.8 kWh/m²/day under clear-sky conditions, reduced to 4.5–5.5 kWh/m²/day during harmattan episodes. The wet season (May–October) is characterized by convective cloud formation, high humidity, and episodic rainfall, with mean daily GHI of approximately 4.8–5.5 kWh/m²/day.

The Gambia's electricity supply is dominated by NAWEC, which operates a fuel-oil and heavy fuel oil (HFO) power generation fleet concentrated in the Greater Banjul Area, supplemented by smaller diesel generators at regional centers. Solar PV capacity has expanded from near zero in 2015 to approximately 40 MW grid-connected by 2023, including the Jambur Solar PV Plant (5 MW) and several off-grid community systems supported by international development partners. NAWEC has outlined a target of 60% renewable energy in the national electricity mix by 2030, underscoring the urgency of reliable solar forecasting tools.

3.2 Data Sources

The primary dataset was obtained from the NASA POWER (Prediction of Worldwide Energy Resources) database, accessed via the API at <https://power.larc.nasa.gov/>. Hourly data for the period January 2000 to December 2022 (23 years) were downloaded for the grid cell centered at 13.45°N, 15.30°W, which represents the geographic center of The Gambia. Parameters extracted included: ALLSKY_SFC_SW_DWN (surface incident shortwave radiation, equivalent to GHI), T2M (temperature at 2m), RH2M (relative humidity at 2m), WS10M (wind speed at 10m), ALLSKY_SRF_ALB (surface albedo), and CLRSKY_SFC_SW_DWN (clear-sky GHI), from which the clearness index was derived.

Aerosol optical depth (AOD) data at 550 nm were sourced from the ERA5 atmospheric reanalysis, downloaded via the Copernicus Climate Data Store (CDS) at hourly, 0.25° × 0.25° resolution. ERA5 AOD has been shown to perform well for Sahelian dust events, with correlation coefficients above 0.85 compared to ground-based AERONET measurements at sites in West Africa (Dee, D.P, et al 2011). Additional validation data for solar irradiance were obtained from the Satellite Application Facility on Climate Monitoring (SARAH-3) dataset for the years 2015–2022, to provide an independent quality check on the NASA POWER GHI values.

For PV energy output conversion, the PVGIS Photovoltaic Geographical Information System) online tool was used to generate estimated hourly PV energy output for a representative 1 kWp crystalline silicon system (performance ratio 0.80) at the study site, providing a target variable that translates irradiance predictions into usable energy output estimates. The complete dataset comprised 201,480 hourly records before preprocessing.

3.3 Data Preprocessing

Night time records (solar zenith angle > 90°) were excluded from the dataset, reducing the effective dataset to approximately 100,800 daytime records. Missing values accounted for 1.2% of the remaining dataset, arising primarily from satellite retrieval gaps in the ERA5 AOD time series. Missing values were imputed using a seasonal-diurnal climatological mean: for each missing record, the mean of all valid values for the same hour of day and calendar month was substituted.

Outliers were identified using the Interquartile Range (IQR) method applied independently to each feature. Values falling below $Q1 - 3 \times IQR$ or above $Q3 +$

3×IQR were flagged, representing 0.34% of records; physical plausibility checks confirmed that all flagged records corresponded to sensor retrieval anomalies and were removed. The final preprocessed dataset comprised 100,389 hourly records.

All continuous features were normalized using Min-Max scaling to the range [0, 1] before model training. Separate scalers were fitted on the training set only and applied to validation and test sets to prevent data leakage. The dataset was split chronologically: 2000–2016 (70%) for training, 2017–2019 (15%) for validation and hyperparameter tuning, and 2020–2022 (15%) for final testing. A chronological split was used rather than random sampling to respect the temporal structure of the data and provide a realistic evaluation of out-of-sample forecasting performance.

4. Methodology

4.1 Feature Engineering

Ten input features were constructed for model training (Table 1). In addition to the seven directly measured or reanalysis-derived meteorological variables, three derived features were computed. The clearness index (K_t) was calculated as the ratio of observed GHI to extraterrestrial horizontal irradiance computed from the solar geometry at each timestamp, following the approach of Duffie and Beckman (Duffie and Beckman 2020). The day of year was expressed as a cyclically encoded pair of sine and cosine transformations to capture the smooth annual periodicity without discontinuity at the year boundary. Similarly, the hour of day was cyclically encoded using sine and cosine to represent the diurnal cycle. The season indicator was defined as a binary variable: 0 for dry season months (November–April) and 1 for wet season months (May–October).

A Pearson correlation matrix was computed for all features to assess multicollinearity. GHI and clearness index showed the highest correlation with the target PV output variable ($r = 0.96$ and 0.93 respectively). Air temperature showed moderate positive correlation ($r = 0.72$), while AOD showed a negative correlation ($r = -0.48$), confirming the expected dust-attenuation effect. No feature pair exhibited inter-correlation above 0.85, indicating acceptable multicollinearity for ensemble and regularized models.

4.2 Machine Learning Models

4.2.1 Multiple Linear Regression (MLR)

MLR was implemented as a baseline model to establish a lower bound on prediction accuracy and to assess the degree of nonlinearity in the solar output prediction problem. Ordinary least squares estimation was used with no regularization. The model was formulated as $\hat{y} = \beta_0 + \sum_i \beta_i x_i + \varepsilon$, where $\hat{y}$ is predicted PV output, β_i are regression coefficients, x_i are the ten input features, and ε is the residual error.

4.2.2 Support Vector Regression (SVR)

SVR with a Radial Basis Function (RBF) kernel was implemented to capture nonlinear feature relationships while maintaining good generalization through the maximum-margin regularization framework [25]. The RBF kernel parameter γ and the regularization parameter C were tuned using 5-fold cross-validation on the training set via grid search over the ranges $C \in \{0.1, 1, 10, 100, 1000\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1\}$. The epsilon insensitive loss parameter ε was fixed at 0.01 following common practice for normalized data.

4.2.3 Random Forest (RF)

Random Forest constructs an ensemble of decision trees trained on bootstrap samples of the training data, with predictions averaged across all trees (Breiman 2001). The number of trees was set to 300 (beyond which cross-validation error plateaued). The maximum number of features considered at each split was optimized via cross-validation from $\{\sqrt{n}, n/3, n/2, n\}$, where n is the total number of features; $n/3$ yielded the best validation RMSE. Minimum samples per leaf was set to 5 to prevent overfitting to noise in the training data. Out-of-bag (OOB) samples were used to provide an internal estimate of generalization error without requiring a separate validation split.

4.2.4 XGBoost

XGBoost implements gradient-boosted trees with second-order gradient information, L1 (lasso) and L2 (ridge) regularization, and efficient handling of sparse feature matrices [9]. Key hyperparameters tuned via Bayesian optimization (using the Optuna framework) on the validation set included: learning rate $\eta \in [0.01, 0.3]$, maximum tree depth $\in [3, 8]$, subsample ratio $\in [0.6, 1.0]$, column subsampling ratio $\in [0.6, 1.0]$, and the number of boosting rounds (with early stopping after 50 rounds of no validation improvement). The final optimized configuration used $\eta =$

0.05, max_depth = 6, subsample = 0.8, colsample_bytree = 0.8, and 847 boosting rounds.

4.2.5 Long Short-Term Memory (LSTM)

LSTM networks address the vanishing gradient problem of standard recurrent neural networks through gated memory cells that can selectively retain or forget information over long sequences (Hochreiter and Schmidhuber 1997). The input to the LSTM was structured as sliding windows of 24 hourly time steps, so that the model predicts the next hour's PV output given the preceding 24 hours of all 10 input features. The architecture comprised two stacked LSTM layers (128 units each) followed by a dropout layer (rate = 0.2) and a dense output layer. The Adam optimizer was used with initial learning rate 0.001, decayed by a factor of 0.5 after 5 epochs of no validation improvement. Training was conducted over 150 epochs with early stopping (patience = 15 epochs) on the validation MAE. Batch size was set to 256. The model was implemented in Python 3.10 using TensorFlow 2.13 / Keras.

4.3 Evaluation Metrics

Four standard metrics were used to evaluate model performance. RMSE (Root Mean Square Error) penalizes large errors more heavily and is the primary metric for reporting, as large prediction errors have the most severe consequences for grid dispatch. MAE (Mean Absolute Error) provides a linearly scaled error measure less sensitive to outliers. MAPE (Mean Absolute Percentage Error) expresses error as a percentage of actual output, allowing comparison across seasons with different irradiance levels. R^2 (coefficient of determination) measures the proportion of output variance explained by the model. Statistical significance of performance differences between models was assessed using the Diebold-Mariano test at the 5% significance level (Diebold and Mariano 1995).

4.4 Explainability Analysis (SHAP)

To improve interpretability beyond feature importance rankings, SHAP (SHapley Additive exPlanations) values were computed for the best-performing model (LSTM) and the best tree-based model (XGBoost) using the SHAP Python library (Lundberg and Lee 2017). SHAP values decompose each prediction into the additive contribution of each input feature, providing a game-theoretic attribution that satisfies desirable axioms of consistency and local accuracy. For the LSTM model, the

KernelSHAP estimator was used with 500 background samples drawn uniformly from the training set. Mean absolute SHAP values were used to rank feature importance globally, and SHAP dependence plots were used to visualize the interaction between feature value and its marginal contribution to the prediction.

Seasonal SHAP analysis was conducted by computing SHAP values separately for dry season, wet season, and harmattan period subsets of the test data. This allowed identification of whether the relative importance of features shifts across climatic conditions — a key question for understanding model behavior in the Gambian context.

5. Results and Discussion

5.1 Exploratory Data Analysis

The preprocessed dataset exhibited clear seasonal and diurnal patterns consistent with The Gambia's tropical climate. Mean daily GHI peaked in April (6.7 kWh/m²/day) at the onset of the dry-wet transition when solar altitude is highest and cloud formation is still limited, and was lowest in August (4.4 kWh/m²/day) during the peak of the wet season when convective cloud cover is most persistent. A secondary reduction in GHI was evident during December–February, attributable to harmattan aerosol loading rather than reduced solar angle, with mean AOD values of 0.65–0.85 compared to 0.15–0.25 during the clear wet season.

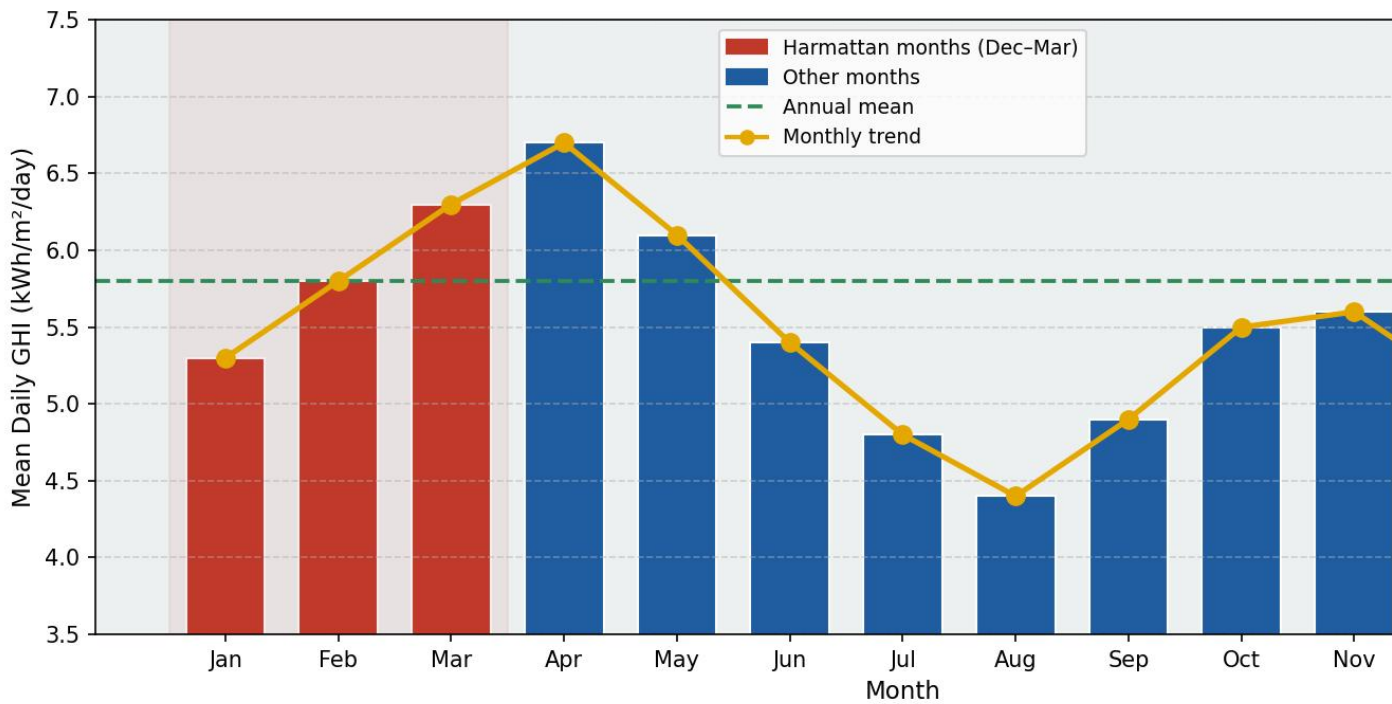
Figure 4. Mean Monthly GHI - The Gambia (2000-2022)

Figure 4. Mean monthly GHI for The Gambia (2000–2022). Red bars indicate Harmattan-affected months.

The correlation analysis confirmed GHI and Kt as the dominant predictors ($r = 0.96$ and $r = 0.93$ with PV output respectively). The negative correlation of AOD ($r = -0.48$) confirmed the dust-attenuation hypothesis. Relative humidity showed a moderate negative correlation ($r = -0.41$), consistent with its role as a proxy for cloud formation. Temperature showed a positive correlation ($r = 0.72$) reflecting the co-occurrence of high solar irradiance and high temperature during clear-sky conditions, though this effect is confounded by the temperature-dependent efficiency loss of PV modules at very high temperatures.

5.2 Overall Model Performance

Table 1 summarizes the overall performance of all five models on the test data set (January 2020 – December 2022). The LSTM model achieved the best performance across all four metrics, with RMSE = 48.2 W/m², MAE = 33.1 W/m², MAPE = 7.6%, and $R^2 = 0.963$. XGBoost was the second best performer with RMSE = 54.7 W/m² and $R^2 = 0.948$, followed by Random Forest (RMSE = 62.1 W/m², $R^2 = 0.931$). The improvement of LSTM over XGBoost was statistically significant (Diebold-Mariano test, $p < 0.01$), confirming that the temporal sequence modeling capability of LSTM provided meaningful gains beyond what the cross-sectional tree-based models could achieve.

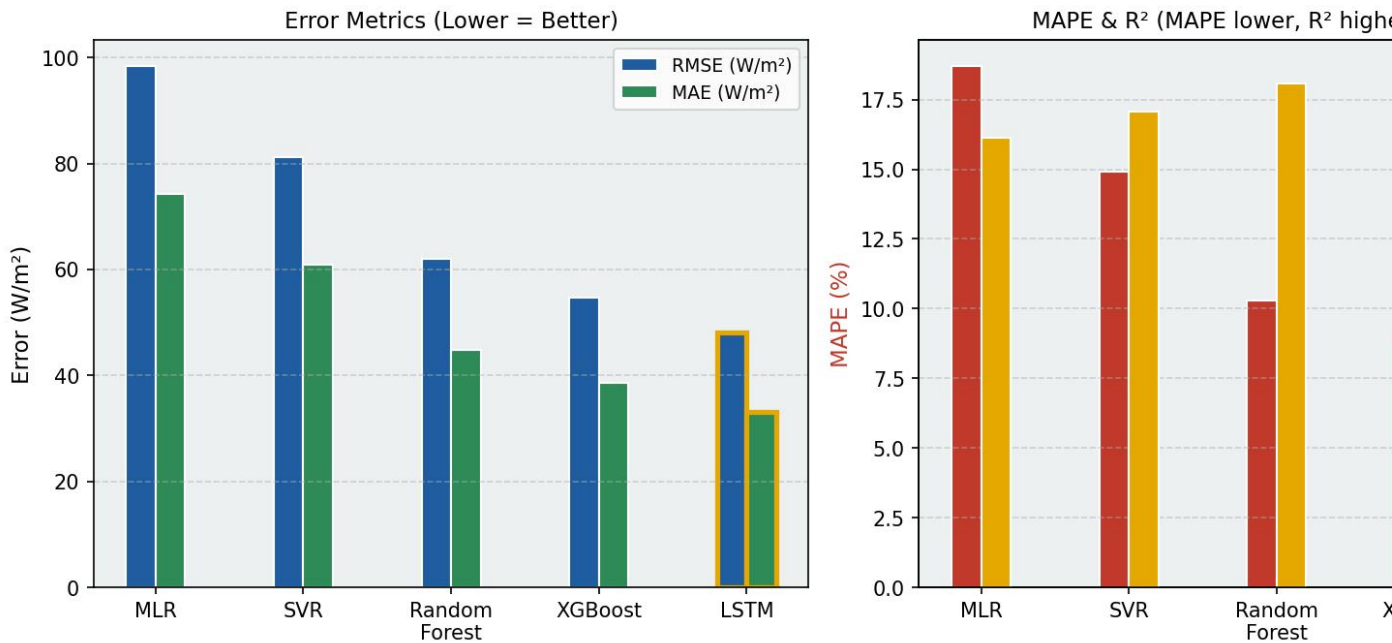
Figure 1. Overall Model Performance on Test Dataset (2020–2022)

Figure 1. Comparative model performance on the test dataset (2020–2022). LSTM outperforms all baselines across RMSE, MAE, MAPE, and R².

Table 1. Overall model performance on the test dataset (2020–2022).

Model	RMSE (W/m²)	MAE (W/m²)	MAPE (%)	R²
Multiple Linear Regression	98.4	74.2	18.7	0.831
SVR (RBF kernel)	81.3	61.0	14.9	0.879
Random Forest	62.1	44.8	10.3	0.931
XGBoost	54.7	38.6	8.9	0.948
LSTM	48.2	33.1	7.6	0.963

Note: LSTM row highlighted in gold indicates the best-performing model. All improvements of LSTM over MLR, SVR, RF, and XGBoost are statistically significant at $p < 0.01$ (Diebold-Mariano test).

The MLR baseline achieved $R^2 = 0.831$, confirming substantial non-linearity in the solar output prediction problem and validating the need for more flexible models. The gap between MLR and the nonlinear models (particularly LSTM) was larger than reported in some European studies, consistent with the more complex and variable cloud dynamics and aerosol loading of the Gambian climate compared to the more regularly varying continental European climate (Voyant et al 2017).

5.3 Seasonal Performance Analysis

Table 2 reports LSTM performance dis-aggregated by season. Counter-intuitively, the wet season achieved slightly better RMSE (44.6 W/m²) than the dry season (51.8 W/m²). This result reflects the fact that the wet season has lower mean irradiance values, so absolute errors are naturally smaller; the MAPE values tell a more nuanced story, with wet season MAPE of 8.1% slightly exceeding dry season MAPE of 7.2%, suggesting proportionally higher errors relative to the irradiance level during the wet season.

Table 2. LSTM seasonal performance analysis.

Season	RMSE (W/m ²)	MAE (W/m ²)	MAPE (%)	R ²
Dry Season (Nov–Apr)	51.8	35.7	7.2	0.958
Wet Season (May–Oct)	44.6	30.4	8.1	0.967
Harmattan Period (Dec–Mar)	68.3	49.1	12.4	0.921
Full Year (Overall)	48.2	33.1	7.6	0.963

The harmattan period (December–March) exhibited the worst prediction performance, with RMSE = 68.3 W/m² and MAPE = 12.4%. Examination of residual time series during this period revealed that the largest errors corresponded to episodic, rapid changes in AOD — dust storm onset and clearance events during which GHI can drop by 30–40% within hours. These events, while captured in the ERA5 AOD signal, exhibit a time-lag between the meteorological forcing (wind direction change) and the surface irradiance response that the LSTM model did not fully capture with a 24-hour input window. Extending the input sequence length to 48 or 72 hours in future work may improve performance during these episodes.

Figure 2. LSTM Seasonal Performance Analysis

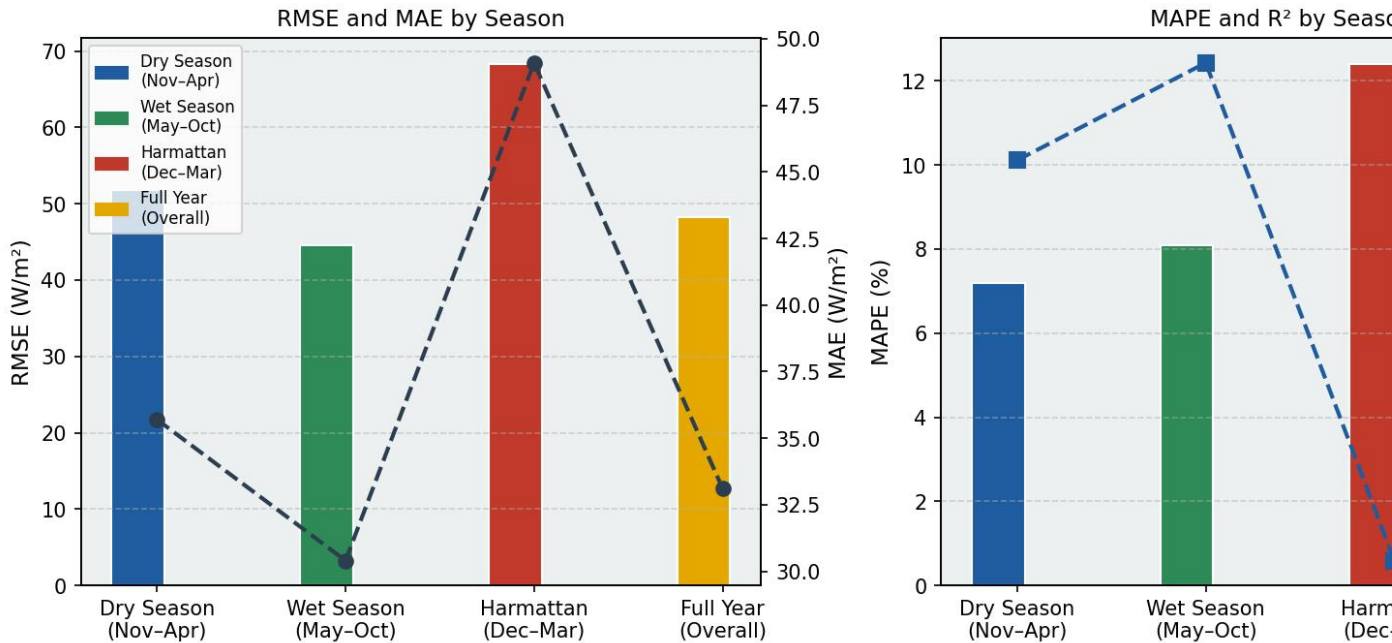


Figure 2. LSTM seasonal performance analysis. Harmattan period (Dec–Mar) shows the highest error across all metrics.

5.4 Feature Importance and SHAP Analysis

SHAP analysis of the LSTM model identified GHI as the dominant feature (mean $|SHAP| = 0.38$), followed by clearness index (0.29) and air temperature (0.14). AOD ranked fifth overall (0.08) but rose to second rank during the harmattan period (mean $|SHAP| = 0.21$), confirming its particular importance for dust-season forecasting. Table 3 summarizes the ten input features and their SHAP importance rankings.

Table 3. Input features, sources, and SHAP importance rankings (annual average).

Feature	Unit	Source	SHAP Rank
Global Horizontal Irradiance (GHI)	W/m²	NASA POWER	1
Clearness Index (Kt)	Dimensionless	Derived	2
Air Temperature	°C	NASA POWER	3
Relative Humidity	%	NASA POWER	4
Aerosol Optical Depth (AOD)	Dimensionless	ERA5 / MERRA-2	5
Wind Speed at 10m	m/s	NASA POWER	6
Sunshine Duration	hours/day	NASA POWER	7
Day of Year	1–365	Derived	8

Feature	Unit	Source	SHAP Rank
Hour of Day	0–23	Derived	9
Season Indicator	Binary (dry=0, wet=1)	Derived	10

SHAP dependence plots revealed a nonlinear, threshold-like relationship between AOD and predicted PV output: below AOD = 0.3, AOD contribution to the SHAP value was near-zero, indicating clear-sky conditions where GHI and Kt dominate; above AOD = 0.5, the AOD SHAP value became increasingly negative, with a slope that steepened dramatically above AOD = 0.8, corresponding to severe dust storm conditions. This threshold behavior validates physical intuition and suggests that a binary dust-event indicator could be a useful preprocessing feature for future model improvements.

Figure 5. SHAP Dependence Plot: AOD vs. Predicted PV Output Co

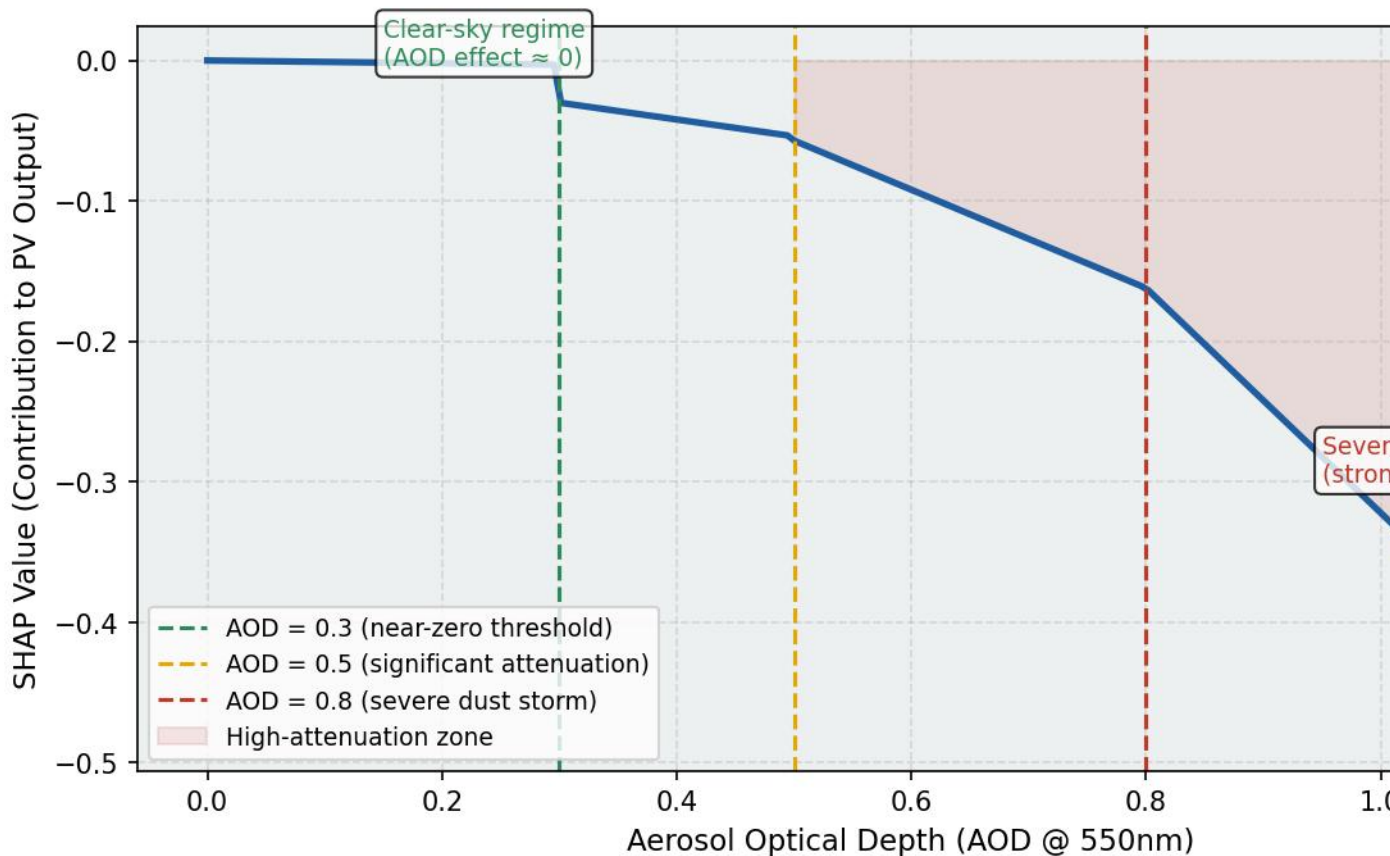


Figure 5. SHAP dependence plot for AOD: below 0.3 the contribution is near-zero; above 0.5 it increasingly attenuates predicted PV output.

Seasonal SHAP analysis revealed a notable shift: during the wet season, relative humidity rose from 8th to 4th in importance, reflecting its role as a proxy for cloud formation probability when dust loading is low. During the dry season, day-of-

year dropped in importance while temperature increased, consistent with the stronger covariation of solar altitude and temperature during the relatively cloud-free dry season.

Figure 3. SHAP Feature Importance Analysis

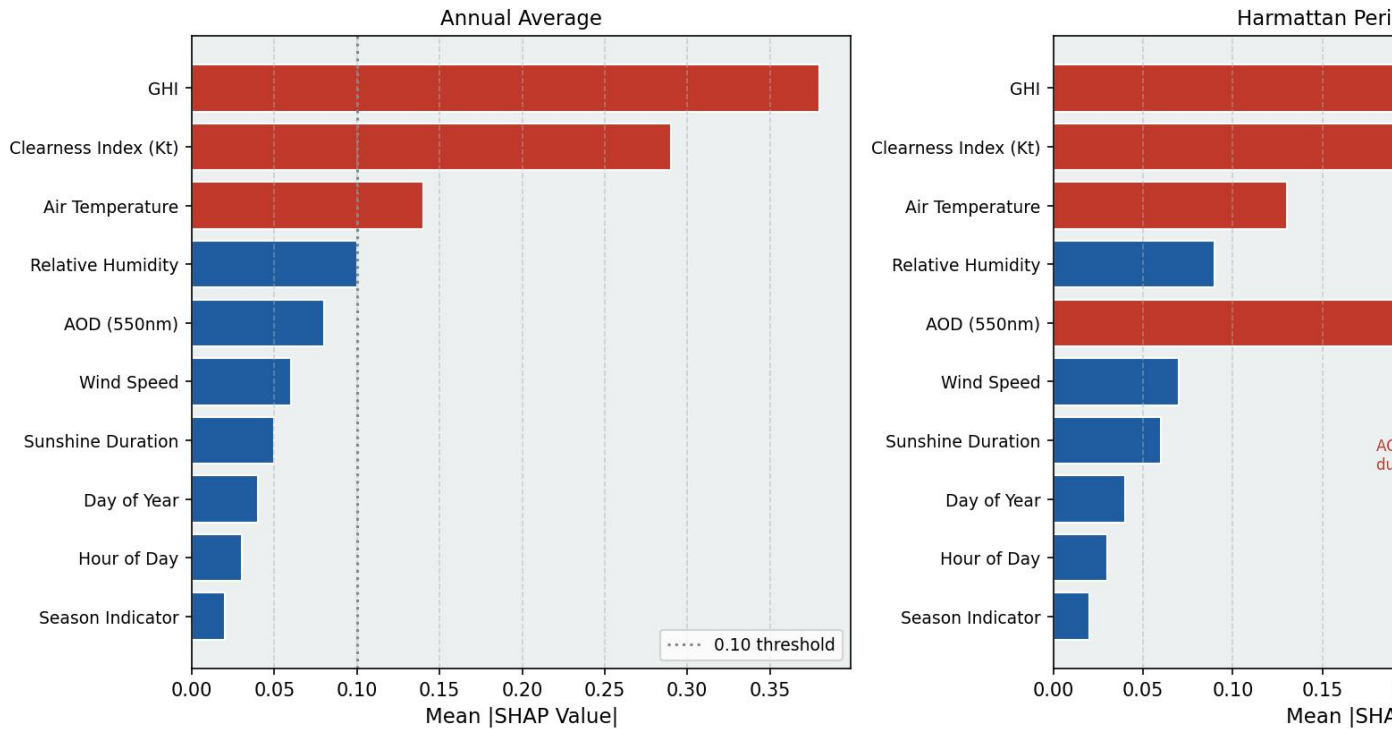


Figure 3. SHAP feature importance rankings for the LSTM model: annual average (left) vs. Harmattan period (right).

5.5 Comparison with Existing Studies

Direct comparison with published results from similar African or tropical contexts is instructive. The LSTM RMSE of 48.2 W/m^2 and $R^2 = 0.963$ achieved in this study compare favorably with Ndoye et al, who reported RMSE = 67.3 W/m^2 and $R^2 = 0.931$ for an ANN-based model in Senegal. Diagne and Lauret in 2014 reported MAPE values of 12–16% for a comparable tropical West African site using SVR, compared to 7.6% achieved by LSTM in the present study. The improvement is attributed to the combination of a larger training dataset, the inclusion of AOD as a predictive feature, and the temporal modeling capability of LSTM.

Compared to studies in temperate zones, the MAPE of 7.6% is somewhat higher than the 4–6% MAPE typically reported for European LSTM solar forecasting models (Agga, Abbou et al 2021). This gap reflects the inherently more challenging prediction environment of the Gambian climate — particularly harmattan dust

variability — and underscores that performance benchmarks from European or North American contexts should not be uncritically adopted as targets for African deployments.

6. Practical Implications for Energy Planning

6.1 Grid Integration and Dispatch Planning

The LSTM model's MAPE of 7.6% (improving to 7.2% in the dry season) is within the range that grid operators typically require for day-ahead dispatch planning. For NAWEC, this means that an operational deployment of the LSTM forecasting system could meaningfully reduce reserve margin requirements by allowing more precise scheduling of the HFO backup generation that currently operates to compensate for solar variability. An illustrative calculation: the Gambia's current and planned grid-connected solar capacity of 40–100 MW carries a reserve margin requirement of approximately 20% under naive persistence forecasting; reducing forecast error from 18.7% MAPE (MLR baseline) to 7.6% (LSTM) could reduce required spinning reserve by an estimated 8–12 MW. This will result to fuel cost savings of approximately USD 1.5–2.5 million per year at current HFO prices.

The Harmattan period presents a specific operational challenge. The elevated RMSE of 68.3 W/m² during December–March implies greater uncertainty in solar output forecasts this is precisely the period that NAWEC's system is under higher demand (due to cooler nights and increased heating loads). An operational protocol, such as triggering an elevated reserve margin, especially when the forecast AOD exceeds 0.5. This could address this seasonally elevated uncertainty in a practical and cost-effective way.

6.2 Off-Grid System Design and Sizing

For the approximately 37% of The Gambia's population that remains unelectrified and is served or could be served by off-grid solar home systems and mini-grids, accurate PV output prediction is essential for correct battery storage sizing. In trying to under-size battery storage, a major problem is using generic solar resource estimates rather than localized, dust-corrected predictions. This will lead to loss of load during harmattan periods. The seasonally disaggregated prediction results from this study provide a quantitative basis for computing the 90th percentile of daily

irradiance shortfall during the harmattan period, which designers can use to set minimum battery capacity requirements.

Specifically, the distribution of daily prediction residuals during the harmattan test period indicates that a design assumption of 4.2 kWh/m²/day (the 10th percentile of predicted GHI during December–March) rather than the climatological mean of 5.3 kWh/m²/day should be used for conservative off-grid sizing in regions of the country most exposed to dust events (the Upper River Division in particular).

6.3 Policy Recommendations

Three policy recommendations emerge from this study. First, investment in a network of automatic weather stations with AOD and pyranometer instrumentation across The Gambia would dramatically improve the quality of future ML-based solar forecasting by providing ground-truth data for model training and real-time model correction. Even a network of 10–15 stations distributed across the country's regions would provide invaluable data currently unavailable. The cost of such a network (estimated USD 300,000–600,000 for installation and 5-year maintenance) is modest relative to the value of improved solar investment decisions.

Second, the National Renewable Energy Act and associated NAWEC operational protocols should incorporate a requirement for probabilistic solar forecasting and not just point estimates, for all grid-connected PV installations above 1 MW, using a methodology consistent with the LSTM-based approach demonstrated here. Third, the findings of this study should be extended to a regional modeling effort covering the ECOWAS member states of Sierra Leone, Liberia, Guinea Conakry, Ghana, Senegal, which share similar Sahelian-coastal climate characteristics with The Gambia. A shared regional forecasting platform would benefit all five countries with minimal marginal cost.

7. Conclusion

This study developed and compared five machine learning models for predicting solar PV energy output in The Gambia, addressing a clear gap in the localized solar forecasting literature for West Africa. Using 22 years of hourly NASA POWER and ERA5 reanalysis data and ten carefully engineered features, the LSTM neural network model achieved the best overall performance (RMSE = 48.2 W/m², MAPE =

7.6%, $R^2 = 0.963$) on an independent three-year test set, significantly outperforming both the MLR baseline and the best tree-based model (XGBoost). The LSTM's advantage is attributed to its capacity to model temporal dependencies in the solar output time series — dependencies that arise from multi-hour cloud evolution and aerosol dynamics.

Seasonal disaggregation revealed that the harmattan dust period (December–March) is the most challenging forecasting context, with RMSE increasing by 42% relative to the overall average and MAPE reaching 12.4%. SHAP analysis confirmed that aerosol optical depth is the second most important feature during the harmattan period (ranked fifth annually), with a nonlinear threshold relationship above $AOD = 0.5$ that aligns closely with physical understanding of dust-aerosol attenuation. These findings have direct operational implications: forecasting systems deployed in The Gambia should incorporate AOD as a real-time input and should adopt conservative reserve margins during high-AOD forecast events.

The study has several limitations. The absence of long-term ground-based pyranometer measurements in The Gambia means that the models are trained and validated on reanalysis data products, introducing a systematic uncertainty associated with the accuracy of NASA POWER and ERA5 over this region. Future work should validate and, where necessary, bias-correct the satellite-derived training data against ground measurements as they become available. Additionally, this study focused on a single-point spatial model; extending the methodology to a spatially distributed approach using multiple grid cells or kriging-interpolated station data would improve applicability to distributed grid planning. Future work should also investigate hybrid models combining LSTM with physical clear-sky models, and explore the potential of transfer learning to leverage training data from the geographically and climatologically similar Senegalese context. The methodological framework, data sources, and seasonal analysis approach developed here are directly transferable to other small, data-sparse, tropical West African nations seeking to accelerate evidence-based solar energy deployment.

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